Chapter 28

The Core of Coalitional Games

Given a transferable utility game (N, v), there are two key questions that are of fundamental interest: (1) Which coalition(s) will form? (2) How should a coalition divide its value among its members? An answer to the second question has a bearing on the answer for the first question. In this chapter, we start studying the answers to these questions using the notion of *the core*, a central notion in cooperative game theory. We discuss the significance of the core of a TU game with several illustrative examples and present several key results concerning this solution concept.

28.1 The Core: Definition and Examples

The core is a set solution concept, that is, it is a set of payoff allocations each of which could be a possible solution to the given TU game Let $N = \{1, \ldots, n\}$ be a set of players and let (N, v) be a TU game. A payoff allocation $x = (x_1, \ldots, x_n)$ is any vector in $\mathbb{R}^n$. x_i is the utility payoff to player i, $i \in N$. We first define the notions of feasible allocation, individually rational allocation, coalitionally rational allocation, and collectively rational allocation.

Definition 28.1 (Feasible Allocation). *An allocation $x = (x_1, \ldots, x_n) \in \mathbb{R}^n$ is said to be feasible for a coalition C if*

$$\sum_{i \in C} x_i \leq v(C)$$

If an allocation x is feasible for C, the players in C can achieve their respective allocations in x by dividing among themselves the worth $v(C)$ that they would be able to earn by cooperating together.

Definition 28.2 (Rational Allocations). *A payoff allocation $x = (x_1, \ldots, x_n)$ is said to be individually rational if*

$$x_i \geq v(\{i\}) \; \forall i \in N$$

It is said to be collectively rational if

$$\sum_{i \in N} x_i = v(N)$$

It is said to be coalitionally rational if

$$\sum_{i \in C} x_i \geq v(C) \;\; \forall C \subseteq N$$

Note that coalitional rationality implies individual rationality. We have, in the previous chapter, defined the notion of an *imputation*. It can be seen that an imputation is a payoff allocation that is individually rational and collectively rational. Collective rationality is the same as (Pareto) efficiency.

Definition 28.3 (The Core). *The core of a TU game (N, v), denoted by $\mathbb{C}(N, v)$, is the set of all payoff allocations that are coalitionally rational and collectively rational.*

$$\mathbb{C}(N, v) = \left\{ (x_1, \ldots, x_n) \in \mathbb{R}^n : \sum_{i=1}^{n} x_i = v(N); \;\; \sum_{i \in C} x_i \geq v(C) \;\; \forall C \subseteq N \right\}$$

In other words, the core is the set of all imputations that are coalitionally rational.

Definition 28.4 (Blocking). *We say a coalition C can* improve on *an allocation $x = (x_1, \ldots, x_n) \in \mathbb{R}^n$ if*

$$v(C) > \sum_{i \in C} x_i$$

We also say that the coalition C blocks *allocation x.*

The above definition implies that C can improve upon x iff there exists some allocation y such that y is feasible for C and every player in C can get strictly higher payoff in y than in x (that is $y_i > x_i \; \forall i \in C$).

Definition 28.5 (The Core: An Alternative Definition). *The core of (N, v) is the set of allocations x such that x is feasible for N and no coalition $C \subseteq N$ can improve upon it.*

We can make the following observations on the concept of the core.

Note. If an allocation x that is feasible for N is not in the core, then there would exist some coalition C such that the players in C could all do strictly better than in x by cooperating together and dividing the worth $v(C)$ among themselves.

Note. If an allocation x belongs to the core, then it implies that for each player, a unilateral deviation will not make the player strictly better off. This means x is a Nash equilibrium of an underlying contract signing game.

Note. The core is an appealing solution concept in the light of the assumption that all coalitions can negotiate effectively. Each allocation in the core could be viewed as resulting out of effective negotiations.

Note. The core of a TU game can be empty. If the core is empty, then we are unable to draw any conclusions about the game. At the same time, if the core consists of a large number of elements, then also, we have difficulty in preferring any particular allocation in the core.

Note. The core of a TU game can be shown to be convex and compact. Convexity and compactness are two desirable properties satisfied by the core. For a proof of this result, the reader is referred to the book by Maschler, Solan, and Zamir [1].

The Core: Some Examples

Example 28.1 (Divide the Dollar Game). For the Divide the Dollar game (Version 1), $\mathbb{C}(N, v)$ is given by.

$$\left\{(x_1, x_2, x_3) \in \mathbb{R}^3 : x_1 + x_2 + x_3 = 300, x_1 \geq 0, x_2 \geq 0, x_3 \geq 0\right\}$$

The core treats the three players symmetrically and includes all individually rational, Pareto efficient allocations. In Version 2 of the game (where players 1 and 2 by themselves can earn 300 without the involvement of player 3), the core will be

$$\left\{(x_1, x_2, x_3) \in \mathbb{R}^3 : x_1 + x_2 = 300, x_1 \geq 0, x_2 \geq 0, x_3 = 0\right\}$$

If players 1 and 2 can earn 300 together without player 3 or players 1 and 3 can earn 300 together without player 2 (Version 3 of the game), it can be seen that the core consists of just a single element, namely $(300, 0, 0)$.

If the players get a nonzero allocation whenever any two players suggest the same allocation (Version 4 of the game - three person majority game), it can be seen that the core is empty. The following two insights provide a rationale for the emptiness of the core in this case. First, when any player i obtains a positive payoff in a feasible allocation, the other two players must get less than 300 which they would be able to gain by themselves. Second, regardless of what the final allocation is, there is always a coalition that could gain if that coalition gets one final opportunity to negotiate effectively against this allocation. □

Example 28.2 (A Three Player TU Game). This example is taken from [2]. Here, $N = \{1, 2, 3\}$ and $v(1) = v(2) = v(3) = 0; v(12) = 0.25; v(13) = 0.5; v(23) = 0.75; v(123) = 1$. Note that $(x_1, x_2, x_3) \in \mathbb{C}(N, v)$ iff

$$x_1 \geq 0;\ x_2 \geq 0;\ x_3 \geq 0$$

$$x_1 + x_2 \geq 0.25;\ x_1 + x_3 \geq 0.5;\ x_2 + x_3 \geq 0.75;$$

$$x_1 + x_2 + x_3 = 1$$

This would mean $x_1 \leq 0.25;\ x_2 \leq 0.5;\ x_3 \leq 0.75$. The core is depicted in Figure 28.1 and happens to be a trapezoid with vertices at $(0, 0.25, 0.75), (0, 0.5, 0.5), (0.25, 0, 0.75)$, and $(0.25, 0.5, 0.25)$. □

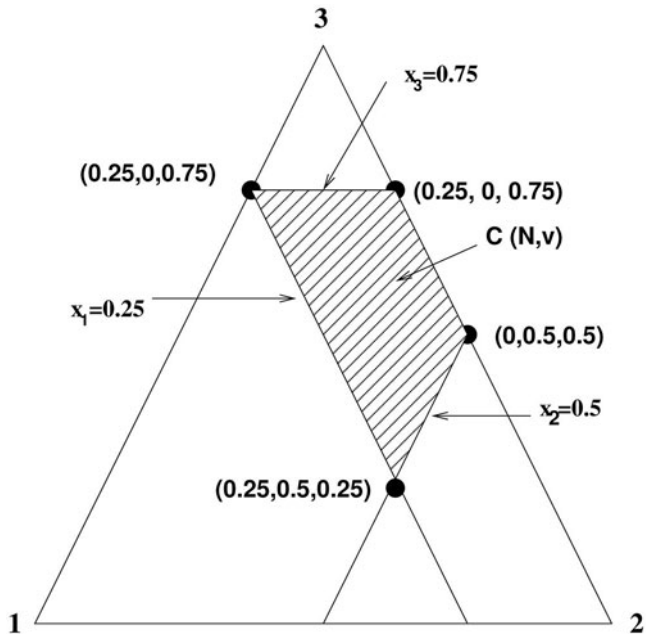

Fig. 28.1: Core of the game in Example 28.2 is a trapezoid

Example 28.3 (House Allocation). This example is originally from the book by von Neumann and Morgenstern [3]. Player 1 has a house which she values at Rs 1 million and wishes to sell the house. There are two potential buyers, player 2 and player 3 who have a valuation of Rs 2 million each and who also have with them Rs 2 million each. Suppose player 1 sells the house to player 2 at a price p million where $1 \leq p \leq 2$. Then the utility of player 1 is p million; utility of player 2 is $(2-p)+2 = 4-p$ million; and the utility of player 3 is 2 million. A similar situation arises if player 1 sells house to player 3 at a price p million, where $1 \leq p \leq 2$. We thus get the characteristic form of the game as:

$$v(1) = 1;\ v(2) = 2;\ v(3) = 2;\ \ v(12) = v(13) = v(23) = 4;\ \ v(123) = 6$$

An allocation $(x_1, x_2, x_3) \in \mathbb{C}(N, v)$ iff

$$x_1 \geq 1;\ x_2 \geq 2;\ x_3 \geq 2;\ \ x_1 + x_2 \geq 4;\ \ x_2 + x_3 \geq 4;\ \ x_1 + x_3 \geq 4;\ x_1 + x_2 + x_3 = 6.$$

The only allocation that satisfies the above equations is $(2, 2, 2)$ which is the lone element in the core. This corresponds to player 1 selling her house at the maximum possible price of Rs. 2 million. □

Example 28.4 (Core of MST Profit Game). Consider the minimum spanning tree game with $N = \{1, 2, 3\}$, discussed in Example 27.7:

$$v(1) = 5;\ v(2) = 9;\ v(3) = 7;\ v(12) = 15;\ v(13) = 12;\ v(23) = 17;\ v(123) = 23$$

The equations defining the core are given by

$$x_1 \geq 5;\ x_2 \geq 9;\ x_3 \geq 7\ x_1 + x_2 \geq 15;\ x_1 + x_3 \geq 12;\ x_2 + x_3 \geq 13;\ x_1 + x_2 + x_3 = 23.$$

The above equations imply that the core is given by the set

$$\{(x_1, x_2, x_3) :\ 5 \leq x_1 \leq 6;\ 9 \leq x_2 \leq 11;\ 7 \leq x_3 \leq 8; x_1 + x_2 + x_3 = 23\}$$

Clearly this is an uncountably infinite set. Note that the core allocation for player 2 is the highest as expected; the core allocation for player 1 is the least which is again according to intuition. □

Example 28.5 (A Glove Market). This example shows that core may be non-empty in some pathological cases. First we consider a simple case and then generalize it. Let there be 5 suppliers of gloves. Of these the first two players can each supply one left glove and the other three players can supply one right glove each. Suppose $N_L = \{1, 2\}$ is the set of left glove suppliers and $N_R = \{3, 4, 5\}$ the set of right glove suppliers. Suppose the worth of each coalition is the number of matched pairs that it can assemble. For example, if $C = \{1, 3\}$, the worth of C is 1, whereas if $C = \{3, 4\}$, the worth of C is 0. In general, given $C \subseteq N$,

$$v(C) = \min\{|C \cap N_L|,\ |C \cap N_R|\}$$

The core of this game can be seen to be the singleton $\{(1, 1, 0, 0, 0)\}$.

On the other hand, if $N_L = \{1, 2, 3\}$ and $N_R = \{4, 5\}$, the core of the game would be the singleton $\{(0, 0, 0, 1, 1)\}$. If the cardinalities of the two sets N_L and N_R are equal, then the core takes a different form altogether. For example, if $N_L = \{1, 2\}$ and $N_R = \{3, 4\}$, the core of the game would contain many elements including $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2})$.

We will now generalize the example to $n = 2,000,001$. Of these, $1,000,000$ players can each supply one left glove and the other $1,000,001$ players can each supply one right glove. The core of this game consists of the unique allocation x such that

$$\begin{aligned} x_i &= 1 \text{ if } i \in N_L \\ &= 0 \text{ if } i \in N_R \end{aligned}$$

The reason for this is:

$$\sum_{i \in N_L} x_i + \sum_{j \in N_R \setminus \{k\}} x_j \ \geq 1,000,000 \quad \forall k \in N_R$$

Suppose that some right glove supplier has some positive payoff in a feasible allocation. Then the total payoff to the other $2,000,000$ players must be less than $1,000,000$, which they could improve by making $1,000,000$ matched pairs among themselves, without this distinguished right glove supplier. An economic interpretation of the example is as follows. Since right gloves are in excess supply, they have a market price of 0. If we add just two more left glove suppliers, the unique core allocation would switch to

$$\begin{aligned} x_i &= 0 \text{ if } i \in N_L \\ &= 1 \text{ if } i \in N_R \end{aligned}$$

in which every left glove supplier gets zero payoff while every right glove supplier gets payoff 1. It would be interesting to see what the core would be if there are $1,000,000$ left glove suppliers and $1,000,000$ right glove suppliers. □

Epsilon Core

The instability displayed by the core for large games (such as the glove market example) can be somewhat overcome using the notion of ε-core or ε-approximate core. Given any number $\varepsilon \geq 0$, an allocation x is in the ε-core of the coalitional game (N, v) iff

$$\sum_{i \in N} x_i = v(N)$$
$$\sum_{i \in C} x_i \geq v(C) - \varepsilon|C| \;\; \forall C \subseteq N$$

If x is in the ε-core, then no coalition C would be able to guarantee all its members ε more than what they get in x.

Example 28.6 (Epsilon Core of Glove Market). For the glove market example, an allocation x is in the ε-core iff

$$\min\{x_i : i \in N_L\} + \min\{x_j : j \in N_R\} \geq 1 - 2\varepsilon$$

This means the worst-off left glove supplier and the worst-off right glove supplier together must get at least $1 - 2\varepsilon$. With $1,000,000$ left glove suppliers and $1,000,001$ right glove suppliers, if $\varepsilon > 0.0000005$, then for any number λ such that $0 \leq \lambda \leq 1$, the allocation x such that

$$x_i = 1 - \lambda \;\; \forall i \in N_L; \quad x_i = \frac{\lambda}{1.000001} \;\; \forall i \in N_R$$

is in the ε-core of the above glove market game. □

28.2 Characterization of Games with Non-Empty Core

The classic, independent findings of Shapley and Bondereva provide a characterization of games with non-empty core, using linear programming [4, 5]. To understand this characterization, we first look into the following linear program (LP), given a TU game (N, v).

$$\text{minimize} \;\; x_1 + x_2 + \ldots + x_n \quad \text{subject to}$$
$$\sum_{i \in C} x_i \geq v(C) \;\; \forall C \subseteq Ni; \;\; (x_1, \ldots, x_n) \in \mathbb{R}^n.$$

The above LP determines the least amount of transferable utility which is necessary for an allocation so that no coalition can improve upon it. Note that this LP definitely has a solution if it is feasible. This is because all the inequalities are of

the *greater than or equal to* type and also because of the structure of the inequalities. Let $(x_1^*, x_2^*, \ldots, x_n^*)$ be an optimal solution of this LP. Then we know

$$\sum_{i \in C} x_i^* \geq v(C) \;\; \forall C \subseteq N$$

In particular, $x_1^* + \ldots + x_n^* \geq v(N)$. There are two possibilities:

(1) $x_1^* + \ldots + x_n^* = v(N)$. In this case, all solutions of this LP will constitute the core $\mathbb{C}(N, v)$. In fact, the core will consist precisely of the solutions of this LP.
(2) $x_1^* + \ldots + x_n^* > v(N)$. In this case, it is clear that the $\mathbb{C}(N, v)$ is empty.

Example 28.7 (Divide the Dollar Game). Recall Version 4 of the divide the dollar game (majority voting game) where $N = \{1, 2, 3\}$ and $v(1) = v(2) = v(3) = 0; v(12) = v(23) = v(13) = v(123) = 300$. The linear program would be:

$$\begin{array}{rc} \text{minimize} & x_1 + x_2 + x_3 \\ \text{subject to} & x_1, x_2, x_3 \geq 0 \\ & x_1 + x_2 \geq 300 \\ & x_2 + x_3 \geq 300 \\ & x_1 + x_3 \geq 300 \\ & x_1 + x_2 + x_3 \geq 300 \\ & x_1, x_2, x_3 \in \mathbb{R} \end{array}$$

The unique optimal solution of this LP is:

$$x_1^* = x_2^* = x_3^* = 150$$

Since $x_1^* + x_2^* + x_3^* = 450 > 300 = v(\{1, 2, 3\})$, the core is empty.

Now consider Version 3 of the Divide the Dollar game. Recall for this game that $N = \{1, 2, 3\}$ and $v(1) = v(2) = v(3) = v(23) = 0; v(12) = v(13) = v(123) = 300$. The LP here is:

$$\begin{array}{rc} \text{minimize} & x_1 + x_2 + x_3 \\ \text{subject to} & x_1, x_2, x_3 \geq 0 \\ & x_1 + x_2 \geq 300 \\ & x_1 + x_3 \geq 300 \\ & x_2 + x_3 \geq 0 \\ & x_1 + x_2 + x_3 \geq 300 \end{array}$$

The unique optimal solution here is:

$$x_1^* = 300; \;\; x_2^* = x_3^* = 0$$

The optimal value is 300. Since $x_1^* + x_2^* + x_3^* = v(N)$, the core consists of a single element $(300, 0, 0)$. $\square$

Example 28.8 (Duals of the Linear Programs for Divide the Dollar). Let us first examine the dual of the LP for the majority voting game. The objective function of the dual LP is:

$$\alpha(1)[v(1)]+\alpha(2)[v(2)]+\alpha(3)[v(3)]+\alpha(12)[v(12)]+\alpha(23)[v(23)]+\alpha(13)[v(13)]+\alpha(123)[v(123)]$$

The dual LP for the majority voting game will be:

$$\begin{aligned} \text{maximize} \quad & 300[\alpha(12)+\alpha(23)+\alpha(13)+\alpha(123)] \\ \text{subject to} \quad & \alpha(1)+\alpha(12)+\alpha(13)+\alpha(123)=1 \\ & \alpha(2)+\alpha(12)+\alpha(23)+\alpha(123)=1 \\ & \alpha(3)+\alpha(13)+\alpha(23)+\alpha(123)=1 \\ & \alpha(1)\geq 0;\ \ \alpha(2)\geq 0;\ \ \ldots;\ \alpha(123)\geq 0 \end{aligned}$$

An optimal solution of this dual LP is given by:

$$\begin{aligned} \alpha(12)=\alpha(13)=\alpha(23)&=\frac{1}{2} \\ \alpha(1)=\alpha(2)=\alpha(3)=\alpha(123)&=0 \end{aligned}$$

Here the core is empty. Now we examine the dual of the LP for Version 3 of the game. The dual LP is:

$$\begin{aligned} \text{maximize} \quad & 300[\alpha(12)+\alpha(13)+\alpha(123)] \\ \text{subject to} \quad & \alpha(1)+\alpha(12)+\alpha(13)+\alpha(123)=1 \\ & \alpha(2)+\alpha(12)+\alpha(23)+\alpha(123)=1 \\ & \alpha(3)+\alpha(13)+\alpha(23)+\alpha(123)=1 \\ & \alpha(1)\geq 0;\ \ \alpha(2)\geq 0;\ \ \ldots;\ \alpha(123)\geq 0 \end{aligned}$$

The optimal value of the dual is again, clearly, 300. The optimal solution is $\alpha(12)=\alpha(13)=\frac{1}{2}$ with all other $\alpha(\cdot)$ values zero. □

Shapley - Bondereva Characterization

In general, the primal LP is:

$$\text{minimize} \sum_{i\in N} x_i \quad \text{subject to}$$

$$\sum_{i\in C} x_i \geq v(C)\ \forall C\subseteq N \quad x_i\in\mathbb{R}\ \ \forall i\in N.$$

The dual LP is given by:

$$\text{maximize} \sum_{C\subseteq N} \alpha(C)v(C) \quad \text{subject to}$$

$$\sum_{C\supseteq\{i\}} \alpha(C)=1\ \forall i\in N;\ \ \alpha(C)\geq 0\ \ \forall C\subseteq N.$$

Let us apply the *strong duality theorem* here. Recall: if the primal (dual) has an optimal solution, then the dual (primal) has an optimal solution and the optimal values are the same. It is easy to see that an optimal solution exists for the above primal. Hence by applying the strong duality theorem, given a TU game (N, v), there exists an allocation $x^* \in \mathbb{R}^n$ and a vector $\alpha^*(C)_{C \subseteq N}$ such that

$$\sum_{i \in C} x_i^* \geq v(C)$$

$$\alpha^*(C) \geq 0 \;\; \forall C \subseteq N$$

$$\sum_{C \supseteq \{i\}} \alpha^*(C) = 1 \;\; \forall i \in N$$

This is illustrated by both Example 28.7 and Example 28.8 presented above. Let us now impose the following condition:

$$\sum_{C \supseteq \{i\}} \alpha(C) = 1 \;\; \forall i \subseteq N \Longrightarrow \sum_{C \subseteq N} \alpha(C) v(C) \leq v(N)$$

That is, feasibility of the dual implies that the value of the objective function of the dual is $\leq v(N)$. We know that the optimal value of the primal $\geq v(N)$. Since the LP has a solution, the optimal solution therefore will have value $v(N)$. This ensures that the core is non-empty. This means we have a necessary and sufficient condition for the non-emptiness of the core. This necessary and sufficient condition is called *balancedness*.

Definition 28.6 (Balanced TU Games). *A TU game (N, v) is said to be balanced if*

$$\sum_{C \supseteq \{i\}} \alpha(C) = 1 \;\; \forall i \in N \Longrightarrow \sum_{C \subseteq N} \alpha(C) v(C) \leq v(N)$$

We are now in a position to state the important theorem that we have just now proved.

Theorem 28.1. *The core of a TU game (N, v) is non-empty iff the game (N, v) is balanced.*

The components of any optimal solution $(x_1^*, x_2^*, \ldots, x_n^*)$ of the above optimization problem are called *balanced aspirations* of the players. The phrase is appropriate since

$$\sum_{i \in C} x_i^* \geq v(C) \;\; \forall C \subseteq N$$

The aspirations are balanced by the fact that $(x_1 + x_2 + \ldots + x_n)$ is minimized.

28.3 The Core of Convex Games

Recall from Chapter 27 that a TU game (N, v) is said to be convex if for every $i \in N$, the marginal contribution of player i to larger coalitions is larger. Interestingly, the core of any convex game is non-empty. The following result on convex games exhibits an allocation that surely belongs to the core of the game.

Proposition 28.1. *Given a convex game (N, v), the allocation $x = (x_1, \ldots, x_n)$ defined by*

$$\begin{aligned} x_1 &= v(\{1\}) \\ x_2 &= v(\{1,2\}) - v(\{1\}) \\ &\ldots \\ x_n &= v(\{1,2,\ldots,n\}) - v(\{1,2,\ldots,n-1\}) \end{aligned}$$

belongs to the core of the game. That is, $x \in \mathbb{C}(N, v)$.

Proof: It can be easily verified for the allocation $x = (x_1, \ldots, x_n)$ defined above, that $x_1 + \ldots + x_n = v(N)$ and therefore X is collectively rational. It remains to be shown that x is coalitionally rational. That is, we have to show that

$$\sum_{i \in E} x_i \geq v(E); \;\; \forall E \subseteq N.$$

Note that $N = \{1, 2, \ldots, n\}$. Suppose E is any coalition of players in N. We can write $E = \{i_1, i_2, \ldots, i_k\}$ for some $k \leq n$. We can assume without loss of generality that $i_1 < i_2 < \cdots < i_k$. This will enable us to observe that

$$\{i_1, i_2, \ldots, i_{j-1}\} \subseteq \{1, 2, \ldots, i_j - 1\}; \;\; \forall j \in \{1, 2, \ldots, k\}.$$

At this point, recall proposition 27.1 which states that a TU game (N, v) is convex iff

$$v(C \cup \{i\}) - v(C) \leq v(D \cup \{i\}) - v(D) \;\; \forall C \subseteq D \subseteq N \;\; \forall i \in N \setminus D.$$

Choosing $C = \{i_1, i_2, \ldots, i_{j-1}\}$; $D = \{1, 2, \ldots, i_j - 1\}$; and $i = i_j \in N \setminus D$, we immediately obtain

$$v(\{i_1, i_2, \ldots, i_j\}) - v(\{i_1, i_2, \ldots, i_{j-1})\} \leq v(\{1, 2, \ldots, i_j\}) - v(\{1, 2, \ldots, i_j - 1\}).$$

Using the above, we next observe that

$$x_{i_1} = (v(\{1, 2, \ldots, i_1\}) - v(\{1, 2, \ldots, i_1 - 1\})) \geq (v(\{i_1\}) - v(\emptyset)),$$

$$x_{i_2} = (v(\{1, 2, \ldots, i_2\}) - v(\{1, 2, \ldots, i_2 - 1\})) \geq (v(\{i_1, i_2\}) - v(\{i_1\})),$$

and so on until

$$x_{i_k} = (v(\{1, 2, \ldots, i_k\}) - v(\{1, 2, \ldots, i_k - 1\})) \geq (v(\{i_1, i_2, \ldots, i_k\}) - v(\{i_1, i_2, \ldots, i_k - 1\})).$$

Summing the above k inequalities and simplifying, we obtain

$$x_{i_1} + x_{i_2} + \cdots + x_{i_k} \geq v(\{i_1, i_2, \ldots, i_k\}) = v(E).$$

The above is clearly true for any $E \subseteq N$, thus proving coalitional rationality. ■

The above proposition immediately leads to the following result which shows that any allocation obtained through a permutation of the set N will also work. We first define some notation. Let $\Pi(N)$ denote the set of all permutations of $N = \{1, 2, \ldots, n\}$. Suppose $\pi \in \Pi(N)$ is any permutation. Let $P(\pi, i)$ denote the set of players who are predecessors of i in the permutation π. That is

$$P(\pi, i) = \{j \in N : \pi(j) < \pi(i)\}.$$

Define the marginal contribution of player i to his predecessors in π by

$$m(P, i) = v(P(\pi, i) \cup \{i\}) - v(P(\pi, i)).$$

Proposition 28.2. *Suppose (N, v) is a convex game and $\pi \in \Pi(N)$ is any permutation. Then the allocation $y^\pi = (m(\pi, 1), m(\pi, 2), \ldots, m(\pi, n))$ belongs to the core of (N, v). That is, $y^\pi \in \mathbb{C}(N, v)$.*

We will be using the above proposition in the next chapter to prove that the Shapley value of any convex game belongs to the core of the game.

28.4 The Core of Cost Games

We have seen in Chapter 27 that TU games where the worth of a coalition is given by the total minimum cost incurred by the players in the coalition are called cost games. When we work with cost games, the definition of the Core will involve $\leq$ instead of $\geq$:

$$\mathbb{C}(N, v) = \left\{ (x_1, \ldots, x_n) \in \mathbb{R}^n : \sum_{i \in C} x_i \leq v(C) \; \forall C \subseteq N; \; \sum_{i \in N} x_i = v(N) \right\}.$$

Example 28.9 (Core of MST cost game). Consider the cost game with $N = \{1, 2, 3\}$ corresponding to the minimum spanning tree game:

$$v(1) = 5; \; v(2) = 1; \; v(3) = 3; \; v(12) = 5; \; v(13) = 8; \; v(23) = 3; \; v(123) = 7$$

The equations defining the core are given by

$$x_1 \leq 5; \; x_2 \leq 1; \; x_3 \leq 3 \; x_1 + x_2 \leq 5; \; x_1 + x_3 \leq 8; \; x_2 + x_3 \leq 7; \; x_1 + x_2 + x_3 = 7.$$

The above equations imply that the core is given by the set

$$\{(x_1, x_2, x_3) : \; 4 \leq x_1 \leq 5; \; -1 \leq x_2 \leq 1; \; 2 \leq x_3 \leq 3; x_1 + x_2 + x_3 = 7\}$$

Note that the core allocation for player 2 could even be negative, meaning player 2 can actually receive money. It would be interesting to compare this with the core of the MST profit game (example 28.4). □

28.5 Summary and References

We summarize here the important concepts discussed in this chapter.

- Given a characteristic form game or TU game (N, v), the core of the game consists of all allocations that are coalitionally rational and collectively rational. Given an allocation $x = (x_1, \ldots, x_n)$ in the core, no coalition C of players will be able to block x because the worth of the coalition, $v(C)$, is less than or equal to the sum of the allocations in x of players of C. The elements of the core can be interpreted as Nash equilibria of an appropriate, underlying contract signing game.
- The core of a TU game can be empty, may be a countable set, or may even be an uncountable set. A game with empty core (majority voting game being an immediate example) implies that negotiations among players may be never ending. The core of a TU game will always be convex and compact.
- If the core consists of a large number of elements, the difficulty would be how to select a particular best element from the core. Other solution concepts in cooperative game theory could be useful here.
- Shapley and Bondereva have provided a characterization of TU games with non-empty core. They have independently proved, using LP duality, that a TU game has non-empty core if and only if it is balanced.
- An appropriate linear programming formulation can be used to determine whether or not a given TU game has non-empty core. The linear program can be used to compute the elements of the core.
- In order to compute the core of a cost game, we need to reverse the inequalities in the core equations from $\geq$ to $\leq$.
- The core of a convex game is non-empty.

The discussion presented in this chapter follows that of Myerson [6] and Straffin [2]. The foundations for the results on the core have been laid in the classic papers by Gerard Debreu and Herbert Scarf [7] and Herbert Scarf [8]. The balancedness characterization of the core is by Bondereva [4] and Shapley [5]. For an excellent detailed treatment of the core and key results concerning the core, we refer the reader to the books by Peleg and Sudholter [9] and Maschler, Solan, and Zamir [1]. In the next chapter, we discuss the Shapley value.

References

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28.6 Exercises

(1) A constant sum TU game (N, v) is one in which

$$v(C) + v(N \setminus C) = c$$

where c is some constant. Show that the core of any essential, constant sum game is empty. (Source: [2]).

(2) Consider the glove market example. What will be the core of this game if there are $1,000,000$ left glove suppliers and $1,000,000$ right glove suppliers?

(3) Consider the following variant of the real estate example. Player 1 has a value of Rs. 1 million; player 2 has value of Rs. 2 million; and player 3 has a value of Rs. 3 million for the house. Player 2 has Rs. 3 million cash, so also player 3. Formulate an appropriate TU game and compute the core. (Source: [2]).

(4) Consider a three person superadditive game with $v(1) = v(2) = v(3) = 0; v(12) = a; v(13) = b; v(23) = c; v(123) = d$ where $0 \leq a, b, c \leq d$. Compute the core for this game. When is the core non-empty for this game? (Source: [2]).

(5) Find the core of the *communication satellites game* defined as follows:

$$v(1) = v(2) = v(3) = 0$$

$$v(12) = 5.2; \;\; v(13) = 2.5; \;\; v(23) = 3; \;\; v(123) = 5.2$$

(Source: [2]).

(6) Consider a TU game (N, v) where $N = \{1, 2, 3\}$ and

$$v(1) = v(2) = 1; \; v(3) = 2; \; v(12) = V(23) = v(13) = 4; \; v(123) = 5.$$

What is the core of this game? (Source: [1]).

(7) Compute the core of the logistics game discussed in Chapter 27. Recall that $N = \{1, 2, 3, 4\}$ and the characteristic function is

$$v(1) = v(2) = v(3) = v(4) = 0$$

$$v(12) = v(13) = v(14) = v(23) = v(24) = v(34) = v(234) = v(123) = 0$$

$$v(134) = 40; \;\; v(124) = 45; \;\; v(1234) = 65$$

(8) Consider a game with five players where player 1 is called a big player and the others are called small players. The big player with one or more small players can earn a worth of 1. The four small players together can also earn 1. Let

$$\begin{aligned} N &= \{1, 2, 3, 4, 5\} \\ v(C) &= 1 \text{ if } 1 \in C \text{ and } |C| \geq 2 \\ &= 1 \text{ if } |C| \geq 4 \\ &= 0 \text{ otherwise} \end{aligned}$$

Compute the core for this game. (Source: [1]).

(9) Let us consider a version of divide the dollar problem with 4 players and total worth equal to 400. Suppose that any coalition with three or more players will be able to achieve the total worth. Also, a coalition with two players will be able to achieve the total worth only if player 1 is a part of the two player coalition. Set up a characteristic function for this TU game and compute the core.

(10) Consider another version of divide the dollar problem with 4 players and total worth equal to 400. Any coalition containing at least two players and having player 1 would be able to achieve the total wealth of 400. Similarly, any coalition containing at least three players and containing player 2 also would be able to achieve the total wealth of 400. Set up a characteristic form game for this situation and compute the core.

(11) It has been stated that the core of a TU game is convex and compact. Prove this result.

(12) A market game is a TU game that consists of a set B of buyers and a set S of sellers such that $N = B \cup S$ and $B \cap S = \emptyset$, and

$$v(C) = \min(|C \cap B|, |C \cap S);\ \forall C \subseteq N.$$

Compute the core of a market game. (Source: [1]).

(13) Consider a TU game with $N = \{1, 2, 3, 4\}$ and $v(C) = 0$ for all coalitions C of cardinality 1 or cardinality 3; $v(C) = 30$ for all coalitions C of cardinality 2; and $v(C) = 50$ for all coalitions C of cardinality 4. Compute the core of this game. (Source: [1]).

(14) **Programming Assignment**. Design a program that, given a TU game, would formulate an appropriate LP and tell us whether the core of the game is empty or non-empty finite or non-empty infinite. In case the core is non-empty and finite, the program should output the elements of the core.